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15 July 1983
SUPERSEDING
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20 September 1974

MILITARY SPECIFICATION

HONING MACHINE, HORIZONTAL, CONNECTING ROD, WITH GAGE; AND CAP AND ROD GRINDER

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a honing machine equipped for resizing journal ends of connecting rods and for pin-hole fitting of rods and pistons for internal combustion engines. The honing machine consists of a precision honing machine and a cap and rod grinder.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-596	-	Connector, Plug, Receptacle, and Cable Outlet, Electrical Power
CC-M-1807	-	Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-LEE-S (R), Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3419

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PPP-D-705 - Drum, Shipping and Storage, Steel (16 and 30 Gallon Capacity)
PPP-D-729 - Drum, Shipping and Storage, Steel, 55 Gallon (208 Liters)

MILITARY

MIL-M-18058 - Machinery, Metal and Wood-Working, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CRF 1910).

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 79 - Electrical Standard for Metalworking Machine Tools

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one complete machine (precision honing machine and cap and rod grinder with all accessories) for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The honing machine and cap and rod grinder shall be of the manufacturer's latest design which shall meet the requirements specified herein. The honing machine shall be designed to hone the journal ends and pin holes of connecting rods and the piston pin holes of pistons. The cap and rod grinder shall be designed to grind a predetermined amount of stock from the parting line area of the cap and rod. The design shall be such that all parts subject to wear, adjustment, and replacement shall be accessible.

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

* 3.4 Construction. The equipment shall be so constructed as to be capable of performing the operations specified herein. All surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free from sharp

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edges. All screws, pins, and similar parts shall be installed to prevent loss of tightness. Parts subject to adjustment or removal shall not be permanently deformed.

* 3.4.1 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking levels.

* 3.4.2 Safety and health requirements. Safety devices shall be provided for all parts of the machine presenting safety hazards. Guards shall be removable to provide easy access to the guarded parts. The machine shall comply with the standards promulgated under OSHA 2206 that are applicable to the machine itself.

3.4.3 Interchangeability. All replaceable parts shall be designed and manufactured to such standards as shall permit replacement and adjustment without the need for modification of the parts or the machine.

* 3.4.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.4.5 Castings and forgings. Castings shall be free of visible blowholes, porosity, hard spots, shrinkage defects, cracks or other defects. Forgings shall be free of visible scale, inclusions, cold shuts, mismatching, or other defects. Mechanical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements of this specification.

* 3.4.6 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations, however, shall not be employed as repair measures for defective parts.

3.4.7 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Rosin base fluxes shall be used in all soldering operations.

3.5 Performance and product characteristics.

* 3.5.1 Honing machine. The honing machine shall be capable of resizing the journal ends of connecting rods within a diameter size range of 1.450 through 3.300 inches by power-stroked honing and a diameter size range of 1.450 through 4.100 inches by manual honing. For pin fitting, the honing machine shall be capable of resizing the pin ends of connecting rods within a diameter size range of 0.720 through 1.400 inches by both power-stroked and manual honing and for bushing expansion within a diameter size range of 0.720 through 1.080 inches by manual expansion. The honing machine shall be capable of honing the journal ends and pin holes of connecting rods and the pin holes of pistons to an accuracy of $\pm$0.0002 inch for roundness within the diameter size ranges specified. The honing machine shall be capable of honing connecting rod bores and pin bores, parallel to each other in both the vertical and horizontal planes within $\pm$0.001 inch per inch of bore length. It shall also be capable of honing piston pin bearing bores (in pistons) perpendicular, within $\pm$0.004 inch, to the longitudinal axis of the piston. The honing machine shall be capable of producing a surface finish of

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16 microinch arithmetical average (aa) or finer. The total time to complete the set of journal ends on the connecting rods shall not exceed 40 minutes, including set-up time, grinding, assembly, and honing.

3.5.1.1 Spindle speed. The honing machine shall have not less than six spindle speed changes within a spindle speed range of 200 through 640 revolutions per minute. Spindle bearings shall be ball or roller type with sealed-in lifetime lubrication.

3.5.1.2 Components.

3.5.1.2.1 Controls. All operating controls shall be conveniently located at the front of the honing machine within the operator's work station.

3.5.1.2.2 Feed dial. The feed dial shall control extension and retraction of the stone(s) in the honing unit and shall limit the maximum stone extension. The feed dial shall allow the stone to feed automatically during the honing operation until the selected pre-set point is reached. The feed dial shall be clearly marked in increments of 0.001 inch and shall be graduated in increments of 0.0001 inch. A feed marker for use in marking feed-up for honing references shall be provided.

3.5.1.2.3 Cutting pressure control. The honing machine shall have an adjustable cutting pressure control to set the cutting pressure of the honing stone against the part being honed. The control shall work automatically and in conjunction with the feed dial and honing dial.

3.5.1.2.3.1 Honing dial. The honing dial shall work in conjunction with the feed dial and shall indicate the action of the honing stone in removing stock from the piece being honed. The honing dial shall be clearly marked in increments of 0.001 inch and shall be graduated in increments of 0.0001 inch.

3.5.1.2.4 Foot control. The honing machine shall be equipped with an adjustable, full machine width pedal bar. Depression of the pedal bar shall actuate the spindle and feed out the stone against the piece being honed. Release of the pedal bar shall stop the spindle and retract the honing stones. The pedal bar's length and tension shall be adjustable for operator comfort and desired pedal pressure. The pedal bar is to be operated directly with a movable pedal by either the right or left foot of the operator. When the pedal bar is depressed and held all the way down, the honing stone shall feed out automatically under the pressure setting on the cutting pressure control and within the limits as pre-set on the feed dial. If the honing machine is belt driven, a belt guard and means for belt tensioning shall be provided.

3.5.1.2.5 Coolant system. The coolant system shall consist of a self-contained honing oil filtering system with a motor-driven filter pump, manifold with shut-off valve and two adjustable independent flow-controlled nozzles, splash guards and oil return pan with a front work adjustment from 14 inches to 26.5 inches, a 10-gallon minimum capacity reservoir, and a baffling system for removing the heavy honing wastes before filtering the coolant. The settled honing wastes shall be accessible for cleaning purposes. The filtering system shall filter all the honing oil at a flow rate of not less than 1.5 gallons per minute. Replaceable filter elements shall be rated for 5-micron filtration. The filtering system shall be wired to operate through the honing machine's electrical system with an "OFF/ON" switch and indicating "ON" light.

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3.5.1.2.6 Base. The base of the honing machine shall be constructed of cast iron or steel. Access shall be provided for maintenance to the coolant system. Provisions shall be made to mount the base to the floor with at least four bolts.

* 3.5.1.2.7 Motors. All honing machine motors shall operate on 230/460 volts, 60 hertz (Hz), three-phase current and shall be initially wired for operation from a 230-volt power source. The spindle motor shall be an induction squirrel-cage type having not less than one horsepower (hp) with continuous duty rating conforming to the requirements of CC-M-1807. The filtering pump motor and the power stroker motor shall be an induction squirrel-cage type having not less than 1/2 hp with continuous duty rating conforming to the requirements of CC-M-1807.

3.5.1.2.8 Gages. The honing machine shall be equipped with a gage, or gages, covering a diameter range of 0.720 to 4.100 inches. The gage(s) shall include a dial-type precision gage with gage point sets for checking pistons and rods for pin fitting, both interference and full-floating type, and for checking journal ends of rods. The dial-type precision gage shall have dial graduations of 0.0001 inch and shall be able to consistently repeat readings to within one-fourth of a dial graduation. The total calibration error of the gage(s) shall not exceed $\pm$0.0005 per revolution of the dial indicator hand.

* 3.5.1.2.9 Work light. The honing machine shall be supplied with a work light, either mounted or unmounted on the machine, to provide illumination at the honing position. The light shall be equipped with not less than two eight-watt fluorescent tubes and shall have a separate switch which shall function independently of the motor switch.

3.5.1.2.10 Mandrels, honing units, and bushing expanders. Mandrels and honing units, with both roughing and finishing stones, and bushing expanders shall be supplied with each honing machine within the diameter ranges for reconditioning connecting rods and piston pin fitting as follows:

(a) Connecting rod journal, 1.450 to 4.100 inches diameter.
(b) Connecting rod bushings, 0.744 to 1.250 inches diameter.
(c) Piston pin fitting, 0.720 to 1.400 inches diameter.
(d) Bushing expansion, 0.720 to 1.080 inches diameter, complete with expander bit and spring, and facing cutter.

Each size mandrel and honing unit in (b) and (c) above shall have a truing sleeve. An adapter sleeve shall be furnished to accommodate the use of different size mandrels and honing units for the machine chuck.

3.5.1.2.11 Power-stroke mechanism. The honing machine shall have an automatic power-stroke mechanism for stroking connecting rod diameters of 1.450 through 3.300 inches. The power-stroking unit shall consist of a drive head to provide various stroke-length ranges, cross beam to provide adjustment of stroke length within each range, arm position knob for adjusting stroke position and for correcting tapers, release handle to disconnect the arm from the stroking power source, connecting rod fixture for holding various sizes of connecting rods, knee support for absorbing honing torque, arm for mounting connecting rod fixture or other work-holding fixtures, setup disk for use in aligning the stroker, indicator support bar for use in aligning or squaring

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squaring the connecting rod fixture, and a rectangular torque bar for controlling the position of the arm. Controls shall be provided to allow manual honing without operating the power-stroke mechanism. A safety switch shall be provided to prevent unintentional operation of the power unit when the arm is up in the manual honing position. Stroke length shall be adjustable from 1/2 to 6 inches and shall have a scale clearly marked and graduated in increments of 1/16 and 1/8 inch. Stroking rates shall be adjustable from 40 to 150 strokes per minute.

3.5.1.3 Accessories. Each honing machine shall include the following:

(a) One diamond stone dresser.
(b) One work support for manual honing.
(c) One V-8 connecting rod support for manual honing.
(d) One work-holding finger-type fixture for power-stroke honing of connecting rod pin ends.
(e) One ring set kit for power-stroke honing. The set shall include all rings in a 1.900- to 3.300-inch range, and with a metal storage case.
(f) Not less than 24 gallons of honing or cooling oil.

3.5.1.4 Tools. All tools, other than common hand tools, required for normal operation and maintenance of the honing machine shall be provided with each machine.

3.5.2 Cap and rod grinder. The cap and rod grinder shall be capable of grinding a predetermined amount of stock from the cap and rod mating surfaces up to a maximum width of three inches and length of six inches. The cap and rod grinder shall be capable of grinding the cap and rod mating surfaces perpendicular to within $\pm$0.002 inch per inch of the vertical axis of the rod and shall produce a surface finish of at least 30 microinches aa. The cap and rod grinder shall be capable of grinding all types of rods, including tongue and groove, except serrated. The grinder shall be furnished with a two-door storage cabinet bench not less than 34 inches high, 36 inches wide, and 24 inches deep. The cabinet shall have nest-type shelves with code tabs for storage of not less than 45 mandrels with sleeves. Shelves shall be tilted to allow honing oil drainage into a bottom trough for easy cleaning. One shelf shall be provided for storage of honing stones, supplies, and accessories.

3.5.2.1 Components.

3.5.2.1.1 Controls. All operating controls shall be mounted for quick and easy access by the operator.

3.5.2.1.2 Micrometer feed. The cap and rod grinder shall include a micrometer feed, requiring one setting to grind a complete set of caps and rods. The micrometer feed shall be calibrated in 0.001-inch increments.

3.5.2.1.3 Face plate and clamp. The cap and rod grinder shall have a hardened steel face plate, locating pin, and a fast action clamping device to lock the cap or rod in place for grinding.

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* 3.5.2.1.4 Motor. Unless otherwise specified (see 6.2), the electric motor shall operate on 115 volts (nominal), 60 Hz, single-phase current. The motor shall be a squirrel-cage type, capacitor-start, continuous duty, and shall not be less than 3/4 hp, totally inclosed (dust free) conforming to the requirements of CC-M-1807.

3.5.2.2 Tools. Tools, other than common hand tools, that are required for normal operation and maintenance of the cap and rod grinder shall be provided, including a diamond wheel dresser having not less than a 0.15 carat diamond.

* 3.6 Electrical components. Electrical components of the honing machine, including relays, switches, wiring, and their inclosures, shall conform to NFPA-79. Belt guard warning labels shall be acceptable in lieu of electrical interlocks. Each belt guard shall be provided with a decal or mylar nameplate with the following warning: "WARNING - Turn off electric power before opening."

3.7 Power supply cord. Each cap and rod grinder shall be furnished with not less than eight feet of three-conductor flexible electrical cord in accordance with NFPA No. 79, type SO, SJO, or STO. The ground conductor shall be connected to the exposed metal parts of the machines to provide a common ground. A three conductor grounding plug shall be provided conforming to style D of W-C-596. A mating three-to-two wire adapter with a green ground conductor shall be furnished with each cord.

3.8 Painting. Unless otherwise specified (see 6.2), painting and finishing of the machine shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall have been cleaned to be free of all foreign matter detrimental to painting, and at least one coat of primer and one coat of enamel, or equivalent, applied.

3.9 Marking. The machine shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified, shall include the National Stock Number.

3.10 Workmanship. Standards of workmanship shall assure that the honing machine shall have the stability, strength, safety, and efficient operating characteristics found in commercial units and as specified in Section 3 herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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* 4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.3.1 First article test conditions. During all first article and initial production tests, the honing machine oil filtering system shall be in operation. All filters used during test will be replaced with new, unused filters before delivery. All applicable equipment in 3.5.1.2.10 and 3.5.1.3, furnished with the machine, shall be used during tests. All honing stones and honing oil used during tests shall also be replaced before delivery. Following the tests, the machine shall be cleaned and prepared for delivery in accordance with Section 5. During tests, all electrical and mechanical controls, including honing dial, shall be inspected and tested to verify compliance with all requirements in Section 3.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (e) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Operational test (see 4.6.1).
(c) Connecting rod honing test (see 4.6.4).
(d) Gage test (see 4.6.5).
(e) Packaging inspection (see 4.7).

* 4.4.1 Sampling. When required (see 6.2), sampling for quality conformance inspection shall be as specified in MIL-STD-105 at the Inspection Level and Acceptable Quality Level (AQL) specified in the contract. For packaging inspection (see 4.7), special inspection level S-4 and an AQL of 4.0 percent defective shall be used.

* 4.5 Product examination. Each complete machine shall be examined to determine compliance with the requirements of Section 3 and the contract. As a minimum, the following characteristics shall be examined:

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Design not as specified	3.2
Material not as specified	3.3
Construction not as specified	3.4
Lubrication	3.4.1
Safety devices	3.4.2
Interchangeability	3.4.3
Threads	3.4.4
Casting and forgings	3.4.5
Welding, brazing, and soldering	3.4.6
Electrical connections	3.4.7
Honing machine components not as specified	
Controls	3.5.1.2.1
Feed dial	3.5.1.2.2

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Cutting pressure control	3.5.1.2.3
Honing dial	3.5.1.2.3.1
Foot control	3.5.1.2.4
Coolant system	3.5.1.2.5
Base	3.5.1.2.6
Motors	3.5.1.2.7
Gages	3.5.1.2.8
Work light	3.5.1.2.9
Mandrels, honing units, and bushing expanders	3.5.1.2.10
Power-stroke mechanism	3.5.1.2.11
Accessories	3.5.1.3
Tools	3.5.1.4
Cap and rod grinder components not as specified	
Controls	3.5.2.1.1
Micrometer feed	3.5.2.1.2
Face plate and clamp	3.5.2.1.3
Motor	3.5.2.1.4
Tools	3.5.2.2
Electrical components not as specified	3.6
Power supply cord not as specified	3.7
Painting and finishing not as specified	3.8
Product identification not as specified	3.9
Workmanship defects	3.10

4.6 Tests.

4.6.1 Operational test. An operational check test shall be conducted on the honing machine for not less than 15 minutes to verify proper operation of all electrical and mechanical controls, motors, and operating mechanisms specified in Section 3 and the contract. The machine shall also be checked for proper belt and pulley alignment, secured hose, pipe, and electrical connections, absence of vibration during all operations, and spindle alignment and run-out in accordance with the manufacturer's specifications.

4.6.2 Capacity and accuracy test. Automotive-type connecting rods of the smallest and largest diameters specified in 3.5.1 shall be set up and honed as follows:

(a) Rod journal ends of 1.450- and 3.300-inch diameters to be honed using the machine's power-stroke mechanism (see 3.5.1.2.11). During this test, the mechanism shall be inspected and tested to verify compliance to all requirements in 3.5.1.2.11, including stroke length and rate and scale graduations.

(b) Rod journal ends of 1.450- and 4.00-inch diameters to be honed using the machine's manual mechanism.

(c) Rod pin ends of 0.720- and 1.400-inch diameters to be honed by both power-stroked and manual honing.

(d) Rod eye bushing expansion of 0.080 and 1.700 inch diameters to be accomplished on the honing machine by manual expansion.

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Accuracy of all finished surfaces, including surface finish, shall be as specified in 3.5.1. After each test in a, b, c, and d above, the specified finished diameters shall be verified using the gage specified in 3.5.1.2.8.

4.6.3 Spindle speed test. The honing machine spindle speed shall be tested to verify compliance with the speed changes and ranges specified in 3.5.1.1.

4.6.4 Connecting rod honing test.

(a) One set, consisting of eight connecting rods of the standard commercial Ford series EBU or B5A6200A, shall be ground on the cap and rod grinder to reduce the journal parting line 0.006 inches. The assembled connecting rod shall then be power-stroked on the honing machine to return the journal end of the rod to the manufacturer's specification. Roundness, straightness, and size of the finished journal end of the connecting rod shall not be more than ±0.0002 TIR of the manufacturer's specifications. Surface finish shall be 16 microinches aa or finer. The total time to complete the set of journal ends on the connecting rods shall not exceed 40 minutes, including set-up time, grinding, assembling, and honing. (See 3.5.1.)

(b) One set, consisting of six connecting rods of the standard commercial Cummins series NH220, shall be ground on the cap and rod grinder to reduce the journal parting line 0.006 inches. The assembled connecting rod shall then be power-stroked on the honing machine to return the journal end of the rod to the manufacturer's specification. Roundness, straightness, and size of the finished journal end of the connecting rod shall not be more than ±0.0003 TIR of the manufacturer's specification. Surface finish shall be 16 microinches aa or finer. The total time to complete the set of journal ends on the connecting rods shall not exceed 30 minutes, including set-up time, grinding, assembling, and honing. (See 3.5.1.)

4.6.5 Gage test. The gage, or gages, specified in 3.5.1.2.8 shall be tested for accuracy and repeatability using calibrated ring gages with certified accuracy traceable to the National Bureau of Standards. A ring gage with an inside diameter of exactly 0.750 inches shall be applied to the gage and the indicator set to 0. The ring gage shall be applied six times without a variation of more than one-fourth dial graduation. A second ring gage having an inside diameter of exactly 0.757 inches shall be applied and the total error shall not exceed ±.0005 inches of the indicated reading. The above tests shall then be repeated using ring gages having inside diameters of exactly 4.000 inches and 4.010 inches.

* 4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking of the complete honing machine shall be Level A (military) in accordance with MIL-M-18058 or Industrial (commercial) in accordance with ASTM-3951-82, as specified (see 6.2). When level A is specified in the contract, the honing or coolant oil shall be stowed in a metal drum(s) conforming to PPP-D-705, type I or II, or PPP-D-729, type I or II, as applicable. The drum(s) of oil may be packed with the honing machine or shipped as an individual container(s).

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6. NOTES

6.1 Intended use. The honing machine covered by this specification provides a precision honing machine and a cap and rod grinder for reconditioning automotive-type connecting rods. The honing machine will resize the journal end bore to accommodate a standard size rod insert and provide for the fitting of a pin to the rod and piston. The reconditioned connecting rod tolerances and fits will conform to the requirements of the rod manufacturer.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Cap and rod grinder motor voltage if different (see 3.5.2.1.4).
(d) Painting and finishing if different (see 3.8).
(e) Product marking if different (see 3.9).
(f) First article inspection if different (see 4.3).
(g) Quality conformance inspection if different (see 4.4).
(h) Sampling when required (see 4.4.1).
(i) Level of preservation, packaging, packing, and marking required (see 5.1).

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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